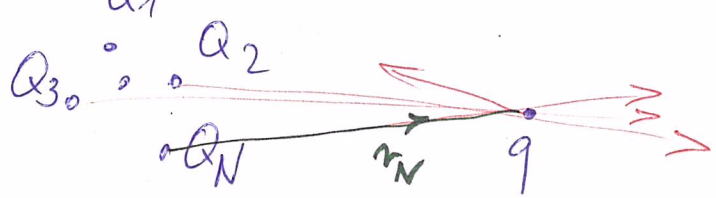
② Gauss's law

2 Electric field is quantized and emanating from point-like objects. (Millikan)

In spite of these facts sometimes it is practical to model ~~el. charg~~ huge quantities of el. charge as a continuum. (analogue to fluid mechanics ☺)

Why: we simply cannot track & calculate the field of $10^{23} - 10^{30}$ interacting point charges.

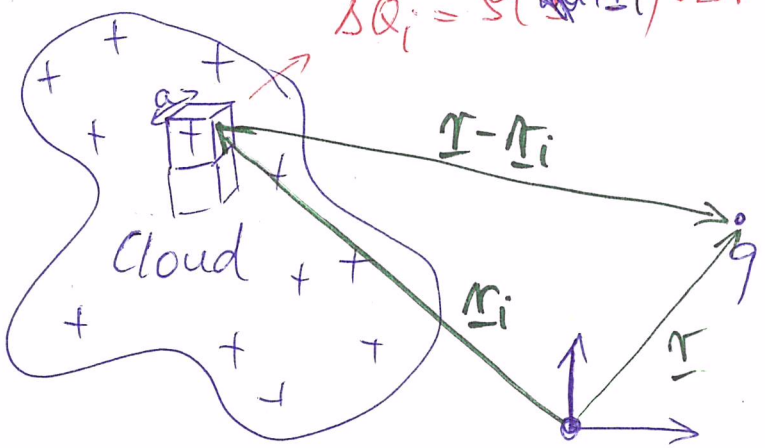
Superposition principle:



$$F_{tot} = \sum_i \underline{F}_i = \frac{1}{4\pi\varepsilon_0} q \sum_i \frac{Q_i}{r_i^2} \underline{e}_i$$

$\underline{E}_i$

$\Delta Q_i = \varrho(\underline{r}_i) \cdot \Delta V$



$$\underline{F}_{tot} \cong \frac{1}{4\pi\varepsilon_0} \cdot q \cdot \sum_i \frac{\Delta Q_i}{|\underline{r} - \underline{r}_i|^2} \cdot \underline{e}_i$$

$$= \frac{1}{4\pi\varepsilon_0} q \sum_i \frac{\varrho(\underline{r}_i) \cdot \Delta V}{|r - r_i|^3} \cdot (\underline{r} - \underline{r}_i$$

$a \to 0$

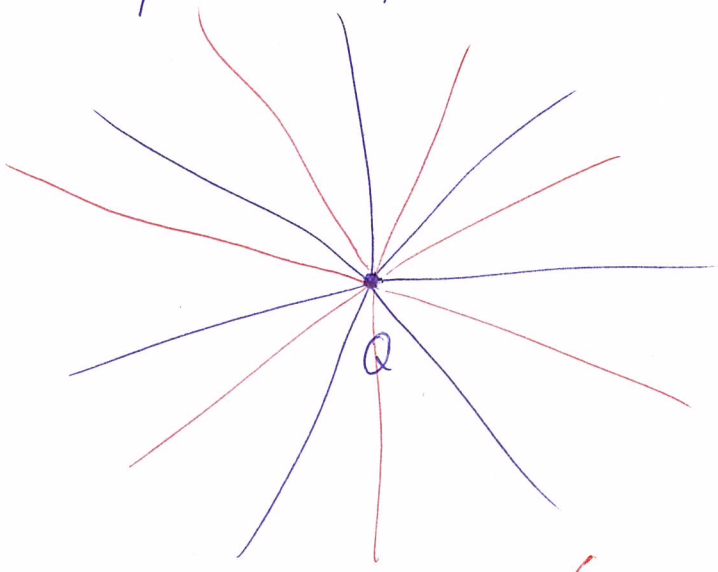
$$\frac{1}{4\pi\varepsilon_0} q \iiint \frac{\varrho(r')}{|r - r'|^3} \cdot (\underline{r} - \underline{r}') \, d^3r' \quad (dV')$$

volumetric

ϱ: charge density (C/m^3)

σ: surface charge density (C/m^2)

Density of Electric field lines $\sim E$



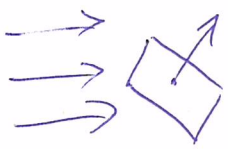
How many field lines should ~~be~~ emanate from Q ?
We can always draw new lines. (Easy in 2D, not so easy in 3D

(Ch 4: Is it possible to make an algorithm to draw an arbitrary N field lines in 3D. What are the N numbers that are easy ? Interlacing ?? possible ?

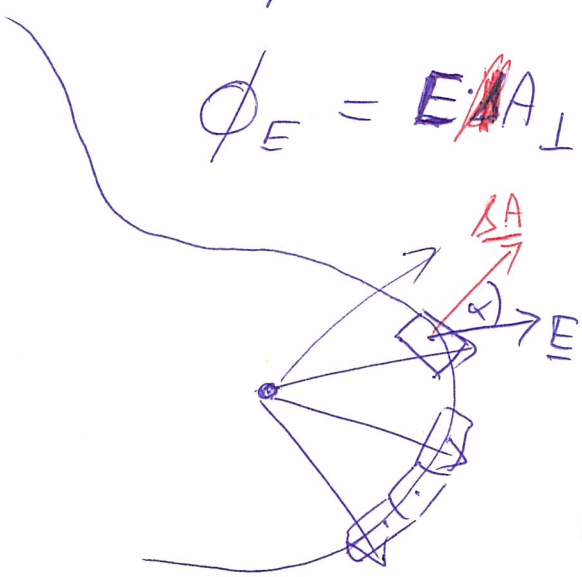
↑

This is not a quantitative physical law, just a proportionality. Let's make it quantitative!

We define the electric flux Φ_E as



$$\Phi_E = E \cdot \Delta A_\perp = E \cdot \Delta A \cdot \cos\alpha = \underline{E} \cdot \underline{\Delta A}$$



In homogeneous field it is easy
In spatially variable field

$$\Delta\Phi_E = \underline{E}(\vec{r}) \cdot \underline{\Delta A}(\vec{r})$$

the total flux is: $\Phi_{E_A} = \sum_{\vec{r}_i} \underline{E}(\vec{r}_i)\underline{\Delta A}_i$

$|\Delta A_i| \to 0$

$\Phi_{E_A} \to \iint_A \underline{E}\, \underline{dA}$ (converges)

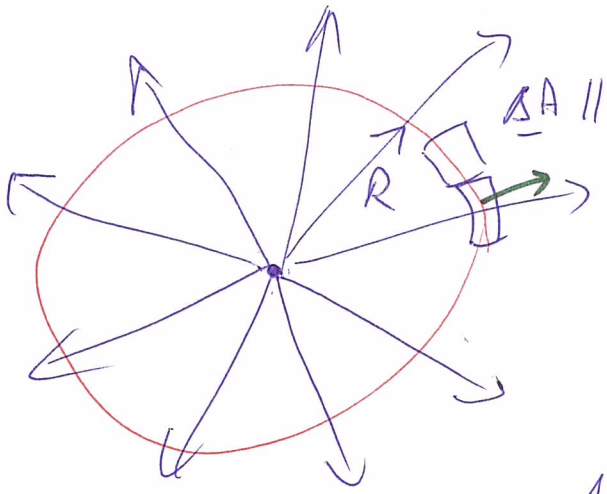
(alternative notation $\int \underline{E} \cdot \underline{ds}$) (Daunbe' cross sectio

Surface integral (other applications? - volume of a flow - electric current)

[$\oint \underline{E} \underline{ds}$: for a closed surface!]

Let's calculate the flux for a point charge!

Easyest: for a concentric sphere!



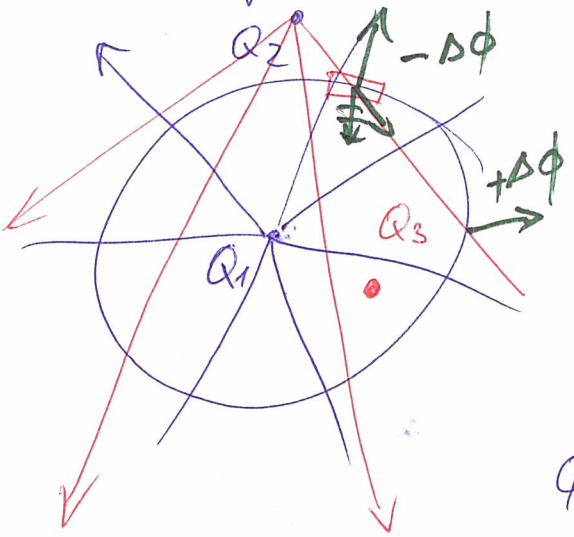
$\underline{\Delta A} \parallel \underline{E}$, $|E|$ is constant

$\Phi_0 = |\underline{E}| \cdot \iint dA \cdot \cos\alpha =$

Coulomb; Surface of sphere

$\Phi_{E(Q)} = \frac{1}{4\pi\varepsilon_0} \frac{Q}{R^2} \cdot 4\pi R^2 = \frac{Q}{\varepsilon_0}$

What if we have a charge outside?



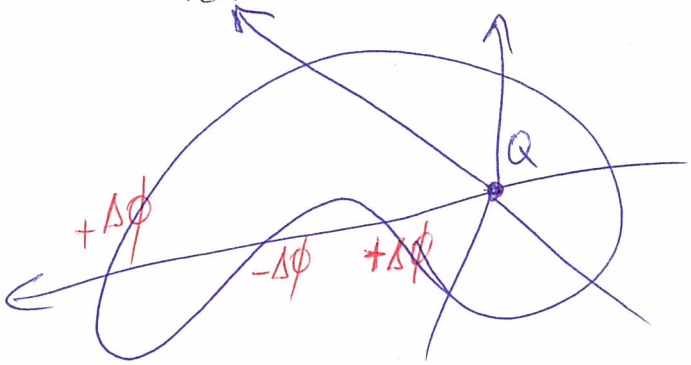
Conventional notation: for a closed surface $\underline{\Delta A}$ (or dA) is always pointing out.

$\Phi_{Q_2 \text{Total}} = \phi$ → position of Q within the sphere is invariant!

$\Phi_{Q_1,Q_3} = \phi_1 + \phi_2$

superposition principle!

shall we take a non-spherical closed surface



the same

Without proof: the flux is ~~~~

Generally valid physical law

$$\Phi_{E_{\text{closed surface}}} = \frac{\sum_{\text{inside}} Q}{\varepsilon_0}$$

$$\oint\oint \underline{E}\, d\underline{s} = \frac{Q}{\varepsilon_0}$$ Gauss's law

(Meaning: electrostatic field is created by charges, Sources of E are charges)

Problem solving: ○, ○—○, // infinite plane

Symmetry of the problem → transferred to the symmetry of the solution!